

HTML Table

The HTML table tag (`<table>`) is used to represent data in a structured way by creating a table. For example,

```
<table border="1" >
  <tr>
    <th>Name</th>
    <th>Age</th>
    <th>Country</th>
  </tr>
  <tr>
    <td>Harry Depp</td>
    <td>28</td>
    <td>Britain</td>
  </tr>
  <tr>
    <td>John Smith</td>
    <td>35</td>
    <td>USA</td>
  </tr>
  <tr>
    <td>Ram Krishna</td>
    <td>19</td>
    <td>Nepal</td>
  </tr>
</table>
```

Browser Output

Name	Age	Country
Harry Depp	28	Britain
John Smith	35	USA
Ram Krishna	19	Nepal

Table tag <table> in HTML

The `<table>` tag is used to define a table. For example,

```
<table>
...
</table>
```

Table Row <tr> in HTML

The `<tr>` tag is used to define a row in a table. For example,

```
<table>
<tr>
...
</tr>
</table>
```

The table row can include either table heading, `<th>` or table data, `<td>`.

```
<tr>
  <th>Name</th>
  <th>Country</th>
</tr>
<tr>
  <td>Prasanna</td>
  <td>Nepal</td>
</tr>
```

```
<tr>
  <td>Simon</td>
  <td>USA</td>
</tr>
```

In a table, there can be any number of rows.

Table Heading, <th> in HTML

The `<th>` tag is used to define a table header. It is generally the top row of the table. For example,

```
<table>
  <tr>
    <th>Item</th>
    <th>Count</th>
  </tr>
  <tr>
    <td>Mango</td>
    <td>125</td>
  </tr>
  <tr>
    <td>Orange</td>
    <td>75</td>
  </tr>
</table>
```

Browser Output

Item	Count
Mango	125
Orange	75

Table Border

Remember we have used the border attribute in our first example.

```
<table border="1">  
  ...  
</table>
```

In HTML, the **border** attribute is used to add a border to a table and all the cells.

In HTML/CSS, **border styles** are used to control the **appearance of borders** around elements like **tables, forms, div, images, etc.**

The border style is defined using the **border-style** property in CSS.

Syntax

```
border-style: style;
```

Example:

```
<p style="border-style: solid;">Hello</p>
```

All Border Styles

Border Style	Description
none	No border
solid	Single straight line
dashed	Dashed line
dotted	Dotted line
double	Double border
groove	3D grooved border
ridge	3D ridged border
inset	Inset border (pressed look)
outset	Outset border (raised look)
hidden	Hidden border

Table Head, Body, and Footer

The HTML table can be divided into three parts: a header, a body, and a footer.

1. Table Header

We use the `<thead>` tag to add a table head. The `<thead>` tag must come before any other tags inside a table. For example,

```
<table>
  <thead>
    <tr>
      <th>Head1</th>
      <th>Head2</th>
    </tr>
  </thead>

  ...
  ...
</table>
```

The content of `<thead>` is placed on the top part of the table and we usually place the rows with table headers inside the `<thead>` tag.

2. Table Body

We use the `<tbody>` tag to add a table body. The `<tbody>` tag must come after `<thead>` and before any other tags inside a table. For example,

```
<table>
  <thead>
    ...
  </thead>
  <tbody>
    <tr>
      <td>Cell 1</td>
      <td>Cell 2</td>
    </tr>
  </tbody>

  ...
  ...
</table>
```

The content of `<tbody>` is placed on the center part of the table and we usually place the rows with the content we want to represent in the `<tbody>`.

3. Table Footer

We use the `<tfoot>` tag to add a table footer. The `<tfoot>` tag must come after `<tbody>` and before any other tags inside a table. For example,

```
<table>
  <thead>
    ...
  </thead>
  <tbody>
    , , , ,
  </tbody>
  <tfoot>
    <tr>
      <td>foot 1</td>
      <td>foot 2</td>
    </tr>
  </tfoot>
</table>
```

The content of `<tbody>` is placed on the bottom part of the table and we usually place the rows with the footer in the `<tfoot>`.

All these tags must be placed inside a `<table>` tag and must contain at least one `<tr>`. For example,

Example: HTML Table Head, Body, and Footer

```
<table>
  <thead>
    <tr>
      <th>S.N</th>
      <th>Item</th>
      <th>Quantity</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>1</td>
      <td>Apple</td>
      <td>2</td>
    </tr>
    <tr>
      <td>2</td>
      <td>Mango</td>
      <td>2</td>
    </tr>
    <tr>
      <td>3</td>
      <td>Orange</td>
      <td>1</td>
    </tr>
  </tbody>
  <tfoot>
    <tr>
      <td></td>
```

```
<td>Total</td>
<td>5</td>
</tr>
</tfoot>
</table>
```

Browser Output

S.N	Item	Quantity
1	Apple	2
2	Mango	2
3	Orange	1
	Total	5

Colspan and

Rowspan

Colspan

The colspan attribute merges cells across multiple columns. For example,

```
<table>
<tr>
  <th>S.N</th>
  <th>Item</th>
  <th>Quantity</th>
</tr>
<tr>
  <td>1</td>
  <td>Apple</td>
  <td>2</td>
```



```

        <td>2</td>
        <td>Mango</td>
        <td>2</td>
    </tr>
    <tr>
        <td>3</td>
        <td>Orange</td>
        <td>1</td>
    </tr>
    <tr>
        <td colspan="2">Total</td>
        <td>5</td>
    </tr>
</table>

```

Browser Output

S.N	Item	Quantity
1	Apple	2
2	Mango	2
3	Orange	1
Total		5

Rowspan

The rowspan attribute merges cells across multiple rows. For example,

```

<table>
  <tr>
    <th>Name</th>
    <th>Subject</th>
    <th>Marks</th>
  </tr>
  <tr>

```

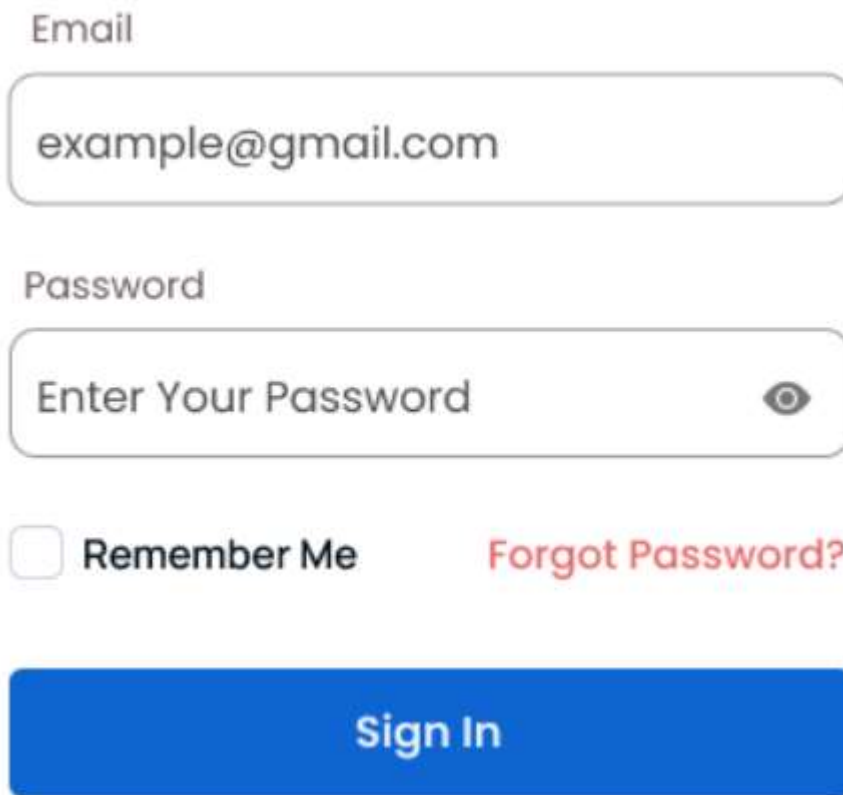
```
<td rowspan="3">Mark Smith</td>
<td>English</td>
<td>67</td>
</tr>
<tr>
<td>Maths</td>
<td>82</td>
</tr>
<tr>
<td>Science</td>
<td>91</td>
</tr>
</table>
```

Browser Output

Name	Subject	Marks
Mark Smith	English	67
	Maths	82
	Science	91

HTML Form

An HTML Form is a section of the document that collects input from the user. The input from the user is generally sent to a server (Web servers, Mail clients, etc). We use the HTML `<form>` element to create forms in HTML.



Email

example@gmail.com

Password

Enter Your Password 

☐ Remember Me [Forgot Password?](#)

Sign In

2.

Label Tag

The `<label>` tag is used to **describe an input field**.

It improves **accessibility and usability**.

Syntax

```
<label>Text</label>
```

Example

```
<label>Name:</label>
```

```
<input type="text">
```

for Attribute

The **for attribute** connects the **label with an input field**.

It must match the **id of the input element**.

Syntax

```
<label for="id_name">Label Text</label>
```

Example

```
<label for="username">Name:</label>
<input type="text" id="username">
```

Now if the user **clicks the label**, the input field will be selected.

id Attribute

The **id attribute** gives a **unique identifier** to an element.

It is used for:

- Connecting **label and input**
- CSS styling
- JavaScript access

name Attribute

The **name attribute** identifies form data when it is sent to the server.

Without **name**, the server **cannot read the value**.

Syntax

```
<input type="text" name="field_name">
```

Example

```
<input type="text" name="username">
```

If a user enters **Hari**, the data sent will be:

```
username = Hari
```

Placeholder Attribute

The **placeholder** attribute shows a **hint inside the input field**.

It disappears when the user starts typing.

Syntax

```
<input type="text" placeholder="Hint text">
```

Example

```
<input type="text" placeholder="Enter your name">
```

Output field will show:

Enter your name

inside the textbox.

Checkbox

What is Checkbox?

A **checkbox** allows the user to **select multiple options** from a list.

Example:

Hobbies → Reading, Music, Sports

User can choose **one or more options**.

Syntax

```
<input type="checkbox">
```

Basic Example

```
<form>
```

Hobbies:


```
<input type="checkbox"> Reading<br>
```

```
<input type="checkbox"> Music<br>
```

```
<input type="checkbox"> Sports<br>
```

```
</form>
```

2. Radio Button

What is Radio Button?

A **radio button** allows the user to **select only one option** from a group.

Example:

Gender → Male / Female / Other

User can select **only one option**.

Syntax

```
<input type="radio">
```

Basic Example

```
<form>
```

```
Gender:<br>
```

```
<input type="radio" name="gender"> Male<br>
```

```
<input type="radio" name="gender"> Female<br>
```

```
</form>
```

⚠ Important:

All radio buttons must have the **same name attribute** to work as a group.

Radio Button with Label

```
<form>
```

```
<label>
```

```
<input type="radio" name="gender" value="male"> Male
```

```
</label><br>
```

```
<label>
```

```
<input type="radio" name="gender" value="female"> Female
```

```
</label>
```

```
</form>
```

4. Checkbox

What is Checkbox?

A **checkbox** allows the user to **select multiple options** from a list.

Example:

Hobbies → Reading, Music, Sports

User can choose **one or more options**.

Syntax

```
<input type="checkbox">
```

Basic Example

```
<form>
```

```
Hobbies:<br>
```

```
<input type="checkbox"> Reading<br>
```

```
<input type="checkbox"> Music<br>
```

```
<input type="checkbox"> Sports<br>
```

```
</form>
```

Here the user can select **any number of options**.

Checkbox with Label and Name

```
<form>
```

```
<label>
```

```
<input type="checkbox" name="hobby" value="reading"> Reading
```

```
</label><br>
```

```
<label>
```

```
<input type="checkbox" name="hobby" value="music"> Music
```

```
</label><br>
```

```
<label>
```

```
<input type="checkbox" name="hobby" value="sports"> Sports
```

```
</label>
```

```
</form>
```

Checkbox with Default Selection

Use **checked** attribute.

```
<input type="checkbox" checked> Reading
```

2. Radio Button

What is Radio Button?

A **radio button** allows the user to **select only one option** from a group.

Example:

Gender → Male / Female / Other

User can select **only one option**.

Syntax

```
<input type="radio">
```

Basic Example

```
<form>
```

```
Gender:<br>
```

```
<input type="radio" name="gender"> Male<br>
```

```
<input type="radio" name="gender"> Female<br>
```

```
</form>
```

⚠ Important:

All radio buttons must have the **same name attribute** to work as a group.

Radio Button with Label

```
<form>
```

```
<label>
```

```
<input type="radio" name="gender" value="male"> Male
```

```
</label><br>
```

```
<label>
```

```
<input type="radio" name="gender" value="female"> Female
```

```
</label>
```

```
</form>
```

Radio Button with Default Selection

Use **checked** attribute.

```
<input type="radio" name="gender" checked> Male
```